Recitation-4:

Q) Study Hour (x_1), Attandance (x_2), Asigment (x_3), Internal score (x_4), Results (y), $\hat{y}_i$, $(y-\hat{y}_i)$

Study Hour (x_1)	Attandance (x_2)	Asigment (x_3)	Internal score (x_4)	Results (y)	$\hat{y}_i$	$(y-\hat{y}_i)$
12	85	80	35	1	0.5	0.5
6	60	55	22	0	0.73	-0.73
15	90	88	40	1	0.5	0.5
5	58	50	20	0	0.73	-0.73
10	75	70	30	1	0.5	0.5
8	65	60	25	0	0.73	-0.73
18	95	92	45	1	0.5	0.5
7	62	24	24	0	0.5	-0.73
14	85	38	38	1		

Find the equation for logistic regression.

Sol:- sigmoid function.

$$\sigma(z) = \frac{1}{1+e^{-z}}.$$

$$z = \omega_1 x_1 + \omega_2 x_2 + \omega_3 x_3 + \omega_4 x_4 + b.$$

$$\omega_1 = \omega_2 = \omega_3 = \omega_4 = 0, \quad b = 0.$$

$$z_1 = 0.$$

$$\sigma(z_1) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-0}} \Rightarrow 0.5.$$

$$\omega^{new} = \omega^{old} - \eta \frac{1}{N} \sum (y - \hat{y}) x_i$$

$$b = b^{old} - \eta \frac{1}{N} \sum_{i=1}^{N} (y - \hat{y})$$

$$J = -\frac{1}{N} \sum_{i=1}^{N} \left[y \log(\hat{y}) + (1-y) \log(1-\hat{y}) \right]$$